1.Introduction

Machine learning models are traditionally trained using centralized datasets, where all data is stored and processed on a single machine. While this approach simplifies training, it introduces challenges related to data privacy, security, and scalability.

Federated Learning (FL) is a distributed machine learning paradigm that allows multiple clients to collaboratively train a shared model without exchanging raw data. Instead of sending data to a central server, each client trains a model locally and only sends model updates to the server. The server then aggregates these updates to produce a global model.

One of the most widely used federated learning algorithms is Federated Averaging (FedAvg). FedAvg allows multiple clients to perform local training and periodically combine their model parameters through weighted averaging.

2. Project Objectives

The primary objectives of this project are:

Implement centralized training using the full dataset.

Implement the FedAvg distributed learning algorithm.

Simulate five clients using multi-process or iterative training.

Train models on:

- FashionMNIST
- CIFAR-10

Measure and compare:

- Training accuracy
- Test accuracy
- Mean accuracy across runs
- Variance of the results
- Analyze the differences between centralized and federated learning.

3. Tools and Technologies

The implementation was performed using the following tools:

Tool	Purpose

Python	Programming language
PyTorch	Deep learning framework
Torchvision	Dataset loading
NumPy	Numerical operations
Matplotlib	Visualization of results

The experiments were executed on a system that automatically selected GPU (CUDA) if available; otherwise, training was performed on CPU.

4. Baseline Centralized Training

In the baseline experiment, the model is trained using all available training data on a single machine.

This represents traditional machine learning training where the dataset is fully centralized.

4.1 Model Architecture

A Convolutional Neural Network (CNN) was implemented using PyTorch. The network consists of:

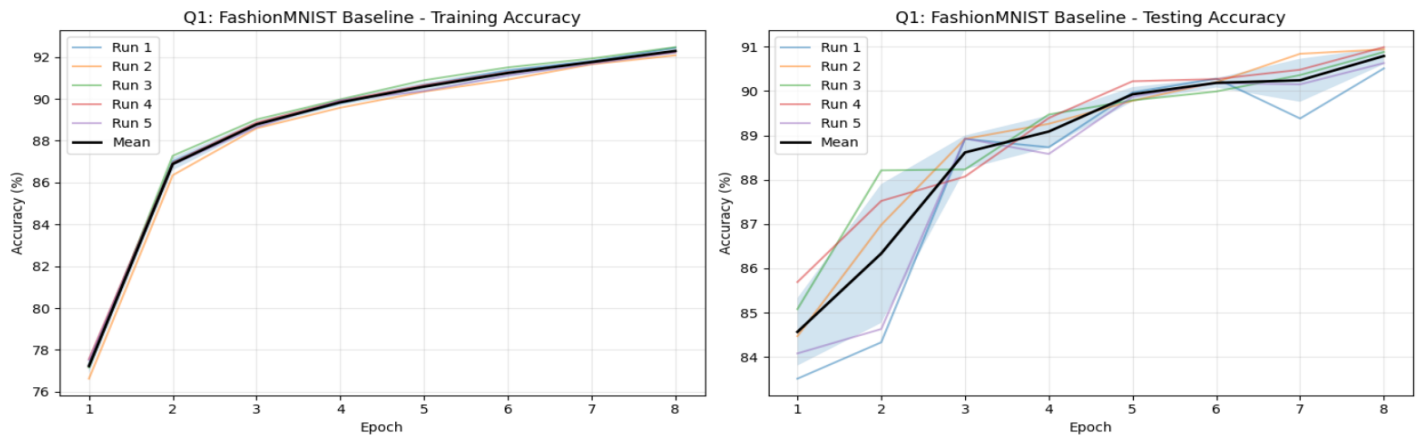
- Convolution layers for feature extraction
- ReLU activation functions
- Pooling layers for dimensionality reduction
- Fully connected layers for classification
- This architecture was intentionally kept small to ensure faster training during experimentation.

4.2 Training Configuration

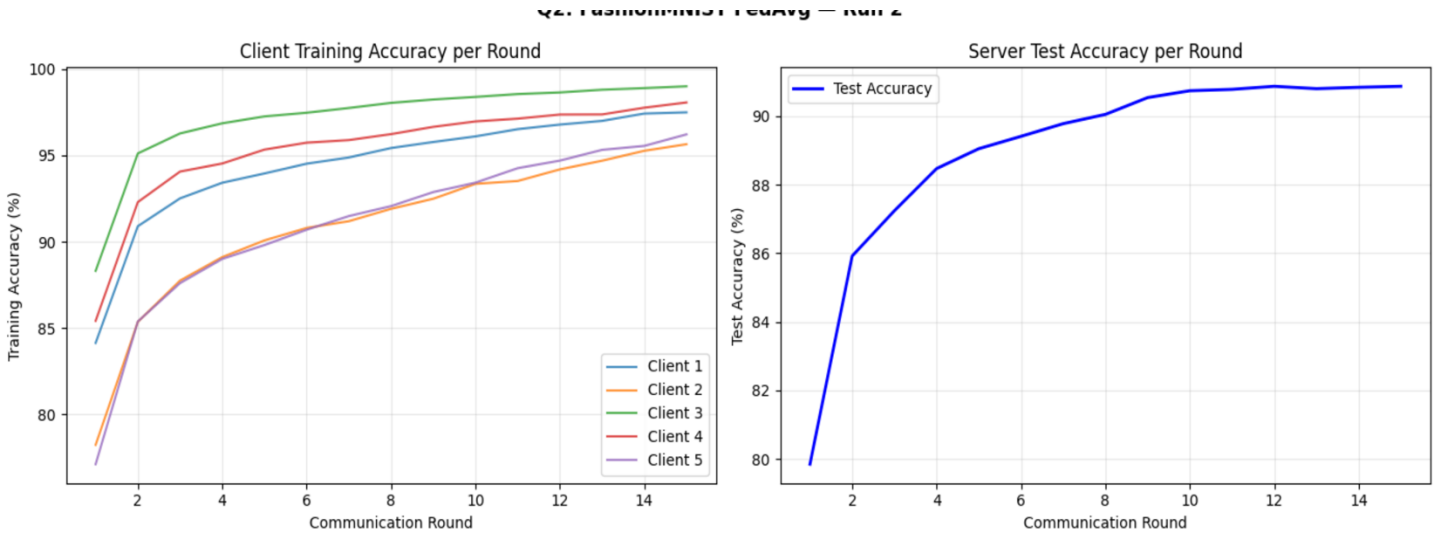
Parameter	Value
Optimizer	SGD
Learning rate	0.01
Batch size	128
Epochs	15
Loss function	Cross-Entropy Loss

Training was repeated multiple times to measure average accuracy and variance.

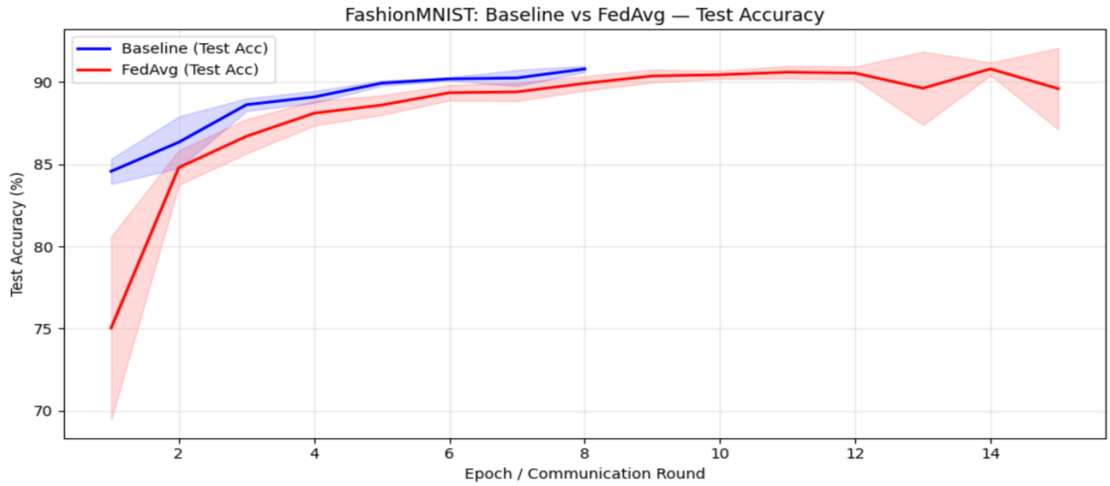
4.3 Training and Testing Accuracy Curves for FashionMNIST Data



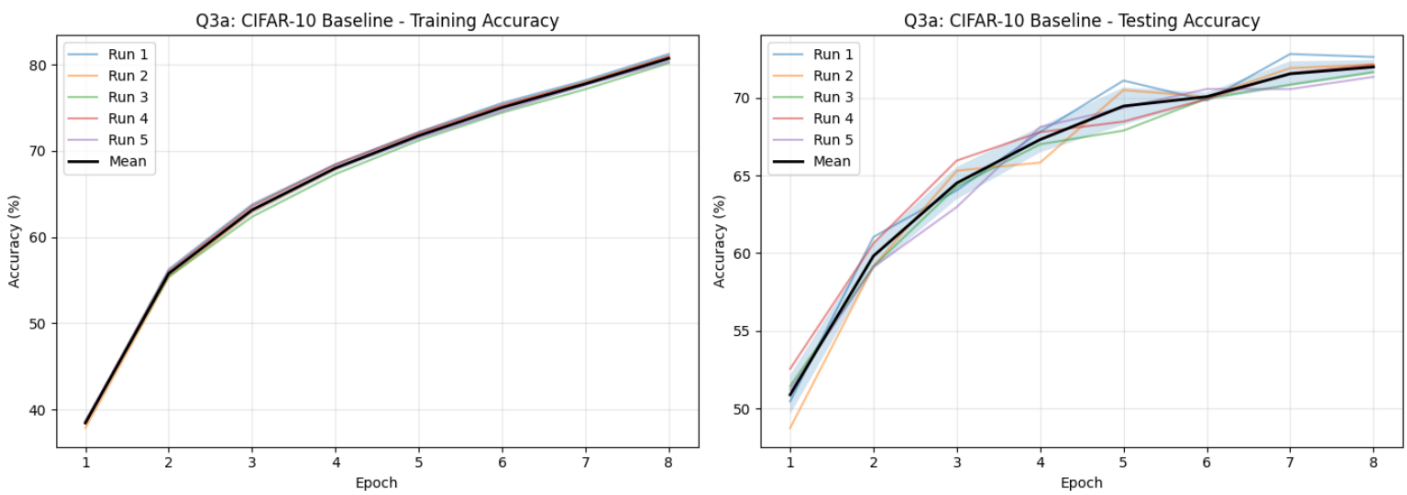

```
Q1: FashionMNIST Baseline — Final Converged Model Results (over 5 runs):
  Training Accuracy:  Mean = 92.30%, Variance = 0.0215
  Testing  Accuracy:  Mean = 90.79%, Variance = 0.0349
```



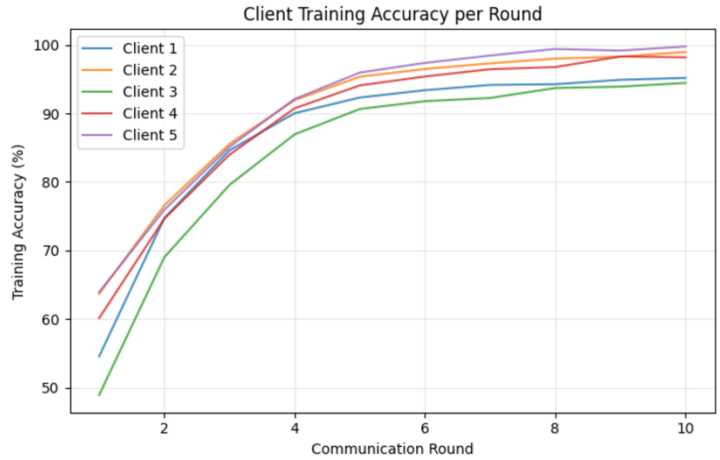
```
============================================================
Comparison: FashionMNIST
============================================================
Method                      Mean Acc (%)    Variance
------------------------------------------------------------
Baseline (Test)             90.79           0.0349
FedAvg (Validation)         89.94           7.2602
FedAvg (Test)               89.59           6.1020
Gap (Test)                  -1.20
```

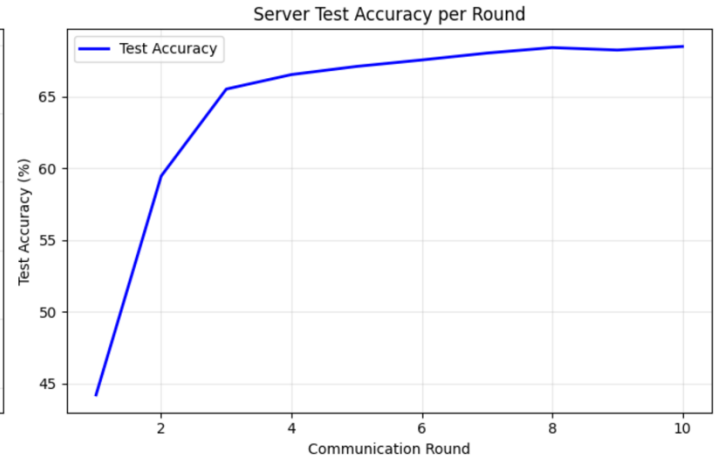


4.4 Training and Testing Accuracy Curves for CIFAR-10 Data

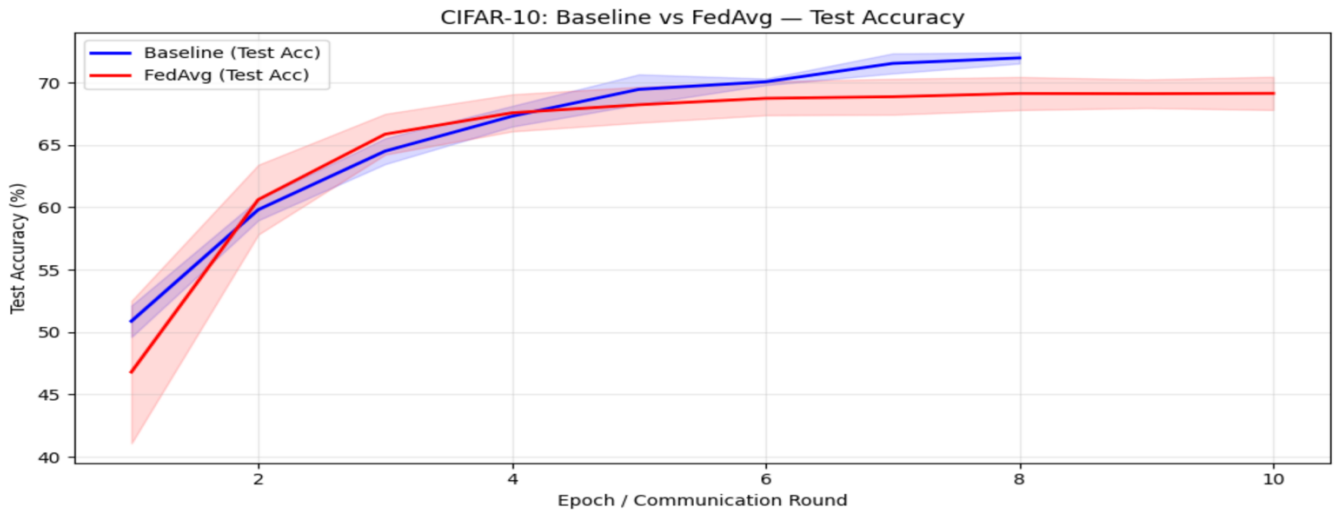



```
Q3a: CIFAR-10 Baseline - Final Converged Model Results (over 5 runs):
  Training Accuracy:  Mean = 80.73%, Variance = 0.1590
  Testing  Accuracy:  Mean = 71.99%, Variance = 0.1952
```






```
Comparison: CIFAR-10
============================================================
Method                      Mean Acc (%)     Variance
------------------------------------------------------------
Baseline (Test)             71.99            0.1952
FedAvg (Validation)         68.97            2.0396
FedAvg (Test)               69.15            1.7698
Gap (Test)                  -2.84
```



5 . Federated Learning Using FedAvg

In the federated learning experiment, the dataset was divided among five simulated clients.

Each client trained the model locally on its own portion of the data.

The central server then aggregated the model parameters using the Federated Averaging algorithm.

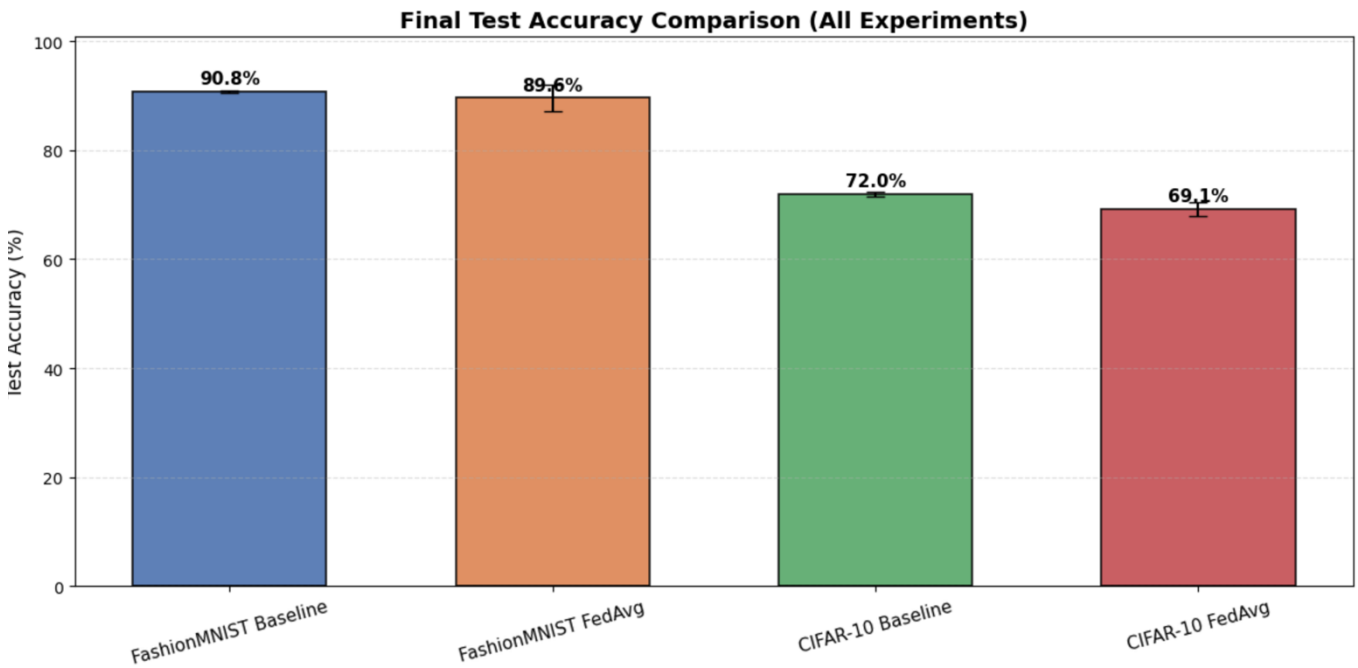
5.1 FedAvg Algorithm

1. The FedAvg process works as follows:
2. The server initializes a global model.
3. The global model is sent to all clients.
4. Each client trains the model on its local dataset.

5. Clients send updated model weights back to the server.
6. The server aggregates these weights using averaging.
7. The updated global model is redistributed to clients.
8. This process repeats for several communication rounds until convergence.

6. Experimental Results

The experiments were repeated several times to compute the mean accuracy and variance.



Method	Mean Accuracy	Std Dev	Variance
FashionMNIST Baseline	90.79%	0.1869	0.0349
FashionMNIST FedAvg	89.59%	2.4702	6.1020
CIFAR-10 Baseline	71.99%	0.4418	0.1952
CIFAR-10 FedAvg	69.15%	1.3303	1.7698

7. Observations

Several important observations can be made from the results.

1. Centralized training achieves slightly higher accuracy

For both datasets, centralized training performed better than federated learning.

- **FashionMNIST:**
  - Baseline: 90.79%
  - FedAvg: 89.59%
- **CIFAR-10:**
  - Baseline: 71.99%
  - FedAvg: 69.15%

This difference occurs because centralized training has access to the entire dataset simultaneously, whereas federated learning aggregates updates from distributed data.

2. Federated learning shows higher variance

The variance of FedAvg results is significantly higher than baseline training.

This occurs because:

- each client trains on different subsets of data
- updates are aggregated periodically
- stochastic training introduces variability

3. CIFAR-10 is significantly harder than FashionMNIST

FashionMNIST achieved approximately 90% accuracy, while CIFAR-10 achieved around 72% accuracy.

This is expected because CIFAR-10 contains complex colored images with higher variability.

4. Accuracy gap increases with dataset complexity

The difference between baseline and federated training was larger for CIFAR-10 compared to FashionMNIST.

This suggests that more complex datasets are more sensitive to distributed training.

8. Challenges Encountered

During implementation, several challenges were encountered:

- Properly splitting datasets among clients

- Synchronizing client training rounds
- Managing model parameter aggregation
- Ensuring reproducibility across experiments
- Managing computational resources for repeated training

9. Conclusion

This project implemented and evaluated the Federated Averaging (FedAvg) algorithm for distributed machine learning.

Experiments were conducted on two datasets: FashionMNIST and CIFAR-10.

The results showed that:

- Centralized training achieves slightly higher accuracy.
- Federated learning introduces higher variance due to distributed data.
- CIFAR-10 is significantly more challenging than FashionMNIST.
- FedAvg remains an effective approach for distributed machine learning despite a small drop in accuracy.

Overall, the project demonstrates the practical implementation of federated learning and highlights the trade-offs between performance and data privacy.

Note: To see the More detailed result please go through the. ipynb file attached with this project